

Short Communication: Analysis of German Gas Storage Drawdown and Supply Risk Assessment (Q1 2026)

Collaborative Agentic Core & Abdelrhim
Energy Data Modeling Team

March 15, 2026

Abstract

This report details the implementation and results of a hybrid predictive model developed to monitor German gas storage levels during the Q1 2026 heating season. By integrating ERA5-Land climate anomalies with industrial cluster-weighted HDD, the model identifies significant storage drawdown trends and supply-side anomalies. Our findings indicate a February 2026 Crisis Delta of 0.1789% and highlight gas price volatility as a primary driver of storage residuals.

1 Introduction

Monitoring gas storage levels is critical for ensuring energy security in Germany. This project utilizes an agentic workflow to develop a high-precision predictive engine capable of detecting deviations from expected drawdown patterns, termed 'Crisis Deltas'.

2 Methodology

The analysis pipeline integrates three primary data streams:

1. Climate Data: Extraction of 2m Temperature and Total Precipitation from ERA5 archives.
2. Industrial Weighting: HDD weighted by industrial cluster density in NRW and SE Germany.
3. Model Architecture: A temporal-lagged XGBoost model optimized for market sensitivity.

3 Results

The model achieved high tracking accuracy (RMSE: 0.0045) across the evaluation period. Anomalous behavior was detected on January 4th, 2026, where the residual exceeded the 5-sigma threshold.

Metric	Value
Root Mean Square Error (RMSE)	0.0045
February 2026 Crisis Delta	0.1789 %
Max Anomaly Magnitude	5.2 sigma

4 Conclusion

The integration of climate anomalies with industrial demand weighting provides a robust framework for energy tracking. Feature importance analysis confirms that gas price volatility remains a critical latent factor in storage drawdown models.